

Confidentiality Concerns -- Guided Notes

Unit tp-6.1 | CompTIA Network+ N10-009 Obj. FC0-U71 6.1 | N10-008 FC0-U71 6.1

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

The CIA triad stands for _____, _____, and _____. These are the three core principles of information security. _____ means that data can only be accessed by authorized individuals. _____ means data is accurate and has not been changed. _____ means systems and data are accessible when needed.

Question 2

Confidentiality means preventing _____ access to information. Threats to confidentiality include _____ (stealing data over a network), _____ (tricking users into revealing credentials), and _____ (collecting discarded documents with sensitive info). The primary countermeasure for confidentiality is _____.

Question 3

Encryption protects confidentiality by converting _____ text into _____ text using an algorithm and key. _____ encryption uses the same key to encrypt and decrypt. _____ encryption uses a public key to encrypt and a private key to decrypt. HTTPS uses _____ to secure web communications.

Question 4

Access control protects confidentiality by restricting _____ to only what each user is authorized to see. _____ access control (MAC) uses security labels set by the system. _____ access control (DAC) lets the resource owner set permissions. _____ access control (RBAC) assigns permissions based on job role.

Question 5

Data classification helps protect confidentiality by labeling data so appropriate _____ controls are applied. Data labeled _____ or _____ should be accessible only to specific authorized individuals. Sensitive data should be _____ in transit (over networks) and _____ at rest (stored on drives).

Question 6

A _____ (VPN) protects confidentiality over public networks by creating an encrypted _____ between the user and the organization. Without a VPN, data sent over public Wi-Fi can be intercepted by _____ attacks where an attacker positions themselves between sender and receiver.

Question 7

The _____ of least privilege states that users and systems should have only the minimum access required for their job. Applying this principle limits _____ damage if credentials are compromised. Over-permissioned accounts are a major source of _____ violations in real breaches.

Question 8 -- Real World Application

REAL WORLD: A healthcare company stores patient records in an unencrypted database. A laptop belonging to an IT employee is stolen from a car. The laptop had a local copy of the database for testing. No patient data is encrypted at rest. Explain what confidentiality failure occurred and describe two controls that would have prevented data exposure.

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/tp-6-1-confidentiality-concerns/tp-6-1-confidentiality-concerns-answer-key.pdf
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